

Introduction to statistics

Parametric and non-parametric tests

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T-tests: summary

T-test in general
Used to compare means

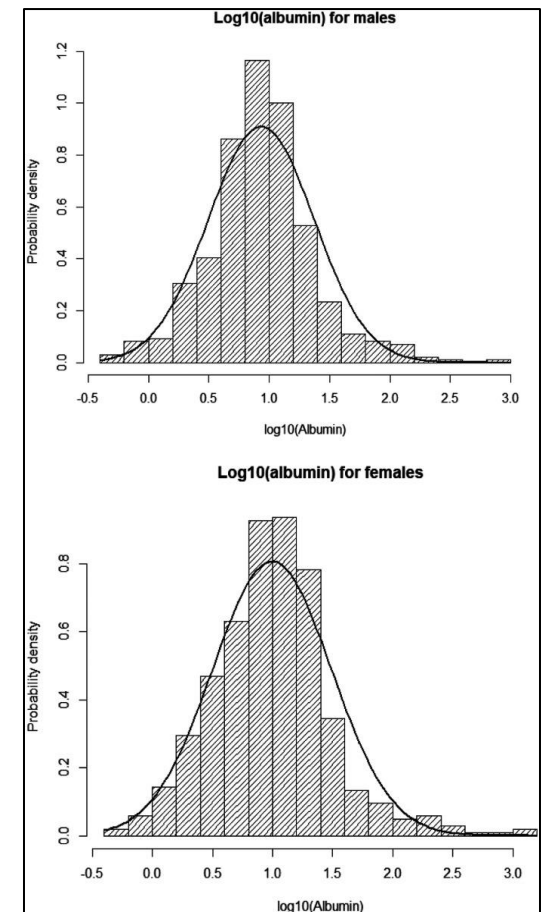
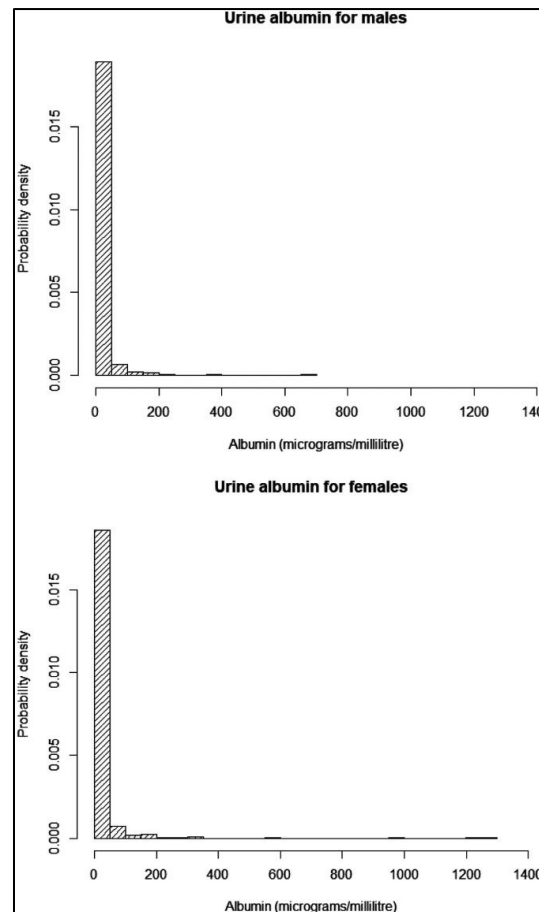
One-sample t-test
Compare the mean of a sample to a given number

Two-sample t-test
Compare the means of two samples

Paired t-test
Compare the difference between pairs of related data points

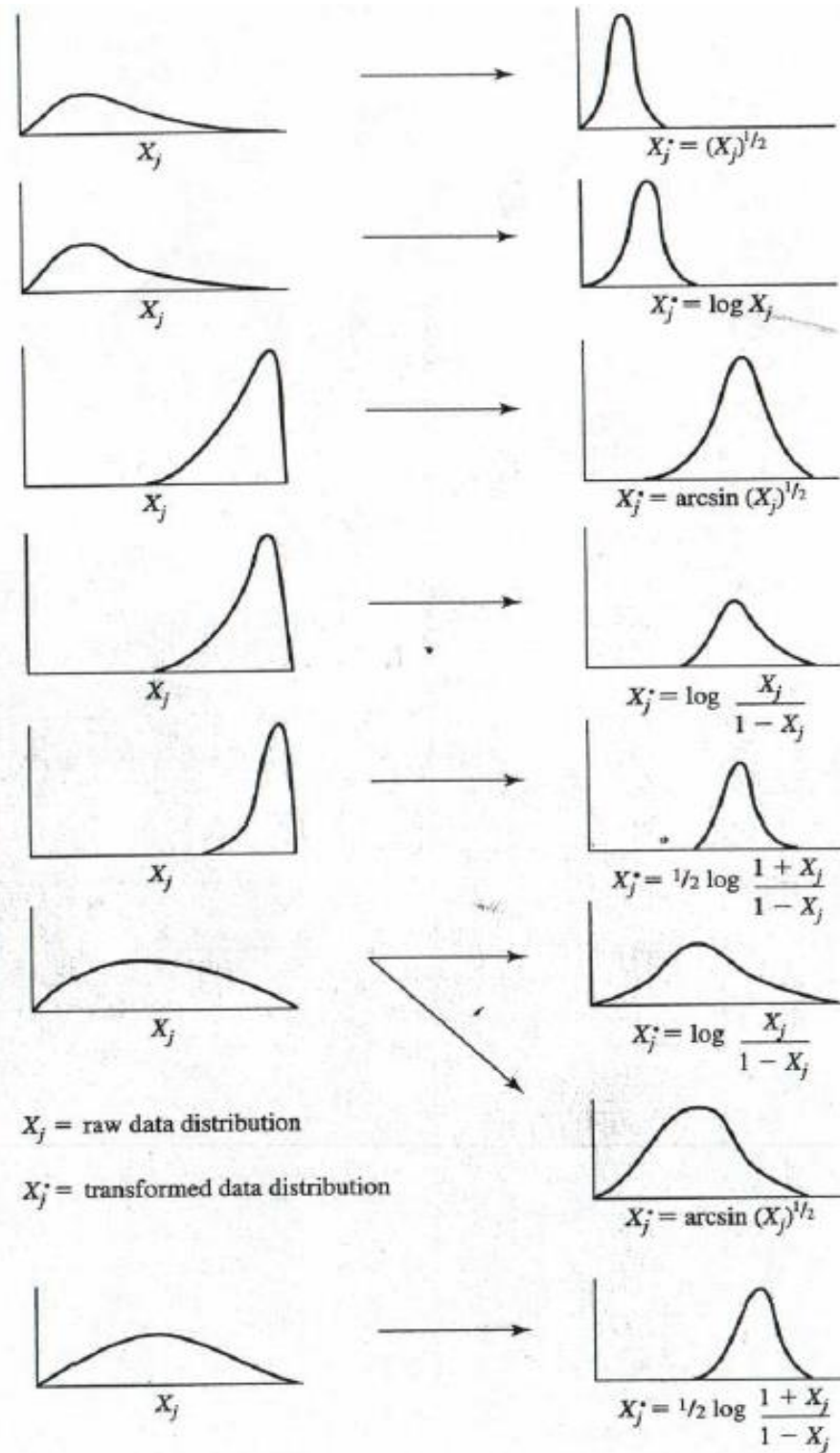
What happens if the data is not normally distributed ?

- Transform the right-tailed data using the log function:
 - Highly asymmetrical data
 - Data spanning several orders of magnitude
 - Data originating from ratios
- Especially important to consider for small sample size cases



<https://pmc.ncbi.nlm.nih.gov/articles/PMC9036143/>

Other transformations for normalisation



What happens if the data is not normally distributed ?

- Data transformations:
 - NB! Your data may now be normal, but interpreting that data may be much more difficult. For example, if you run a t-test to check for differences between two groups → data you are comparing has been transformed, you cannot simply say that there is a difference in the two groups' means. The difference is based on the square root.
- Another possible solution: non-parametric tests
 - Wilcoxon rank-sum test (WRS) == Mann–Whitney U test == Mann–Whitney–Wilcoxon (MWW) (for 2-sample test)
 - Wilcoxon signed-rank test (for paired samples)

Mann–Whitney U test/Wilcoxon rank-sum test (WRS)

This test compares ranks of 2 groups instead of means (2-sample test)

Assumptions:

- All the observations from both groups are independent of each other
- The variable is ordinal or numerical

H0: the distributions of both populations are equal

H1: the distributions of both populations are not equal

In R: `wilcox.test(group1, group2)`

Wilcoxon signed-rank test

This test compares median differences of paired samples instead of means (paired test)

H0: the median difference between the paired samples is zero (no effect).

H1: the median difference between the paired samples is not zero.

Wilcoxon signed-rank test

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H0: the median difference between the paired samples is zero (no effect).

H1: the median difference between the paired samples is not zero.

In R: `wilcox.test(before, after, paired = TRUE)`

If the test statistic, W , is *less than or equal* to the critical value in the table, the null hypothesis can be rejected. Otherwise, we have failed to reject the null hypothesis.

Wilcoxon signed-rank test in practice

Let's say we have a small set of paired data showing the before and after treatment weights of 8 subjects:

Question: Does the treatment help to reduce weight? (→ one-tailed test)

- H_0 : median difference = 0 (or ≥ 0)
- H_1 : median difference < 0

Subject	Before Treatment (X)	After Treatment (Y)
1	72	70
2	85	80
3	90	92
4	76	77
5	88	85
6	94	86
7	102	99
8	75	66

Wilcoxon signed-rank test in practice

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1. Calculate the difference

$$D1 = 72 - 70 = 2$$

$$D2 = 85 - 80 = 5$$

$$D3 = 90 - 92 = -2$$

$$D4 = 76 - 77 = -1$$

$$D5 = 88 - 85 = 3$$

$$D6 = 94 - 86 = 8$$

$$D7 = 102 - 99 = 3$$

$$D8 = 75 - 66 = 9$$

2. Rank the absolute difference (from low to high)

$$D1 = 2 (2.) \rightarrow \text{tied, will be } (2+3) / 2 = 2.5$$

$$D2 = 5 (6.)$$

$$D3 = 2 (2.) \rightarrow \text{tied, will be } (2+3) / 2 = 2.5$$

$$D4 = 1 (1.)$$

$$D5 = 3 (4.) \rightarrow \text{tied, will be } (4+5) / 2 = 4.5$$

$$D6 = 8 (7.)$$

$$D7 = 3 (4.) \rightarrow \text{tied, will be } (4+5) / 2 = 4.5$$

$$D8 = 9 (8.)$$

3. Assign signs based on the differences:

D1: +, D2: +, D3: -, D4: -, D5: +, D6: +, D7: +;

D8: -

4. Sum the positive and negative ranks:

$$\text{Positive ranks: } 2.5 + 6 + 4.5 + 7 + 4.5 = 24.5$$

$$\text{Negative ranks: } 2.5 + 8 = 10.5$$

$$5. \text{ Test statistic } W = \min(\text{abs}(24.5), \text{abs}(10.5)) = 10.5$$

6. Compare W to the critical value: at alpha = 0.05, n = 8, one-tailed, which is 3

$10.5 > 3 \rightarrow$ no sufficient evidence to say that the treatment helps to reduce weight